TITLE–SIMPLE CONSOLE CALCULATOR

CLASS–1 BSC. CS ‘B’

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Introduction

1. This program is a basic console calculator.
2. It allows the user to choose an arithmetic operator (+, -, *, /).
3. The user enters two numbers for the calculation.
4. The program performs the operation using a function.
5. It handles invalid operators and division-by-zero errors.
6. The result of the calculation is displayed to the user.

Objectives

Calculator Program Requirements (C Language)

1. Command-line interface: Program runs in terminal, accepts user input via command-line arguments
2. **Arithmetic functions: Supports** addition, subtraction, multiplication, division of integers/ floats
3. **Input validation:**

   Checks for valid numeric input, non-numeric characters trigger error message
4. **Error handling:** Catches division-by-zero errors, displays "Cannot divide by zero"
5. **Output format:** Results printed with 2 decimal places, clear and readable

Problem Statement

Calculator Function Elements

Problem Statement:
User-defined arithmetic task

Operator Input:
+, −, ×, ÷ selection

Operands: Numeric values
entered by user

Peparate Function:
Square root, percentage, etc.

Result Display: Calculated
value output

Invalid Input: Non-numeric
or unsupported entry

Division by Zero:
Error message prompt

1) Prompts the user to enter an arithmetic operator (+, -, *, or /)

2) Requests two numeric operands from the user

3) Uses a separate function to perform the selected operation

4) Displays the correct result to the user

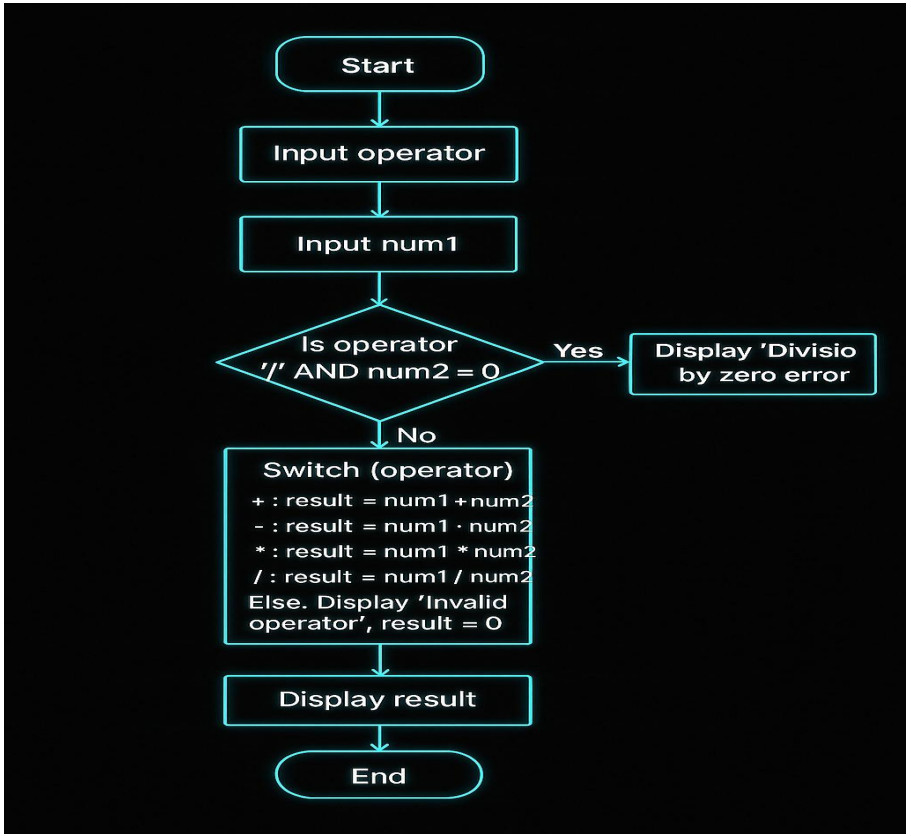
5) Properly handles invalid operator input

6) Prevents division by zero and displays an appropriate error message

Algorithm

```
1. Start
2. Declare variables:
    operator (char)
    num1, num2, result (double)
3. Display message:
    "Enter an operator(+ - * /):
4. Read operator from the user
    "Enter number 1:
5. Read num1 from the user
6. Display message:
    "Enter number 2:
8. If operator is '/'
    and num2 is 0
9. If operator is culate(operator, num1,num2)
  10. If operator is '/' and num2 is 0
      Display: Error: Division by
       zero is undefined.
11.Otherwise, display:
   Result:  followed by result
12.End
```

Flowchart


Start
Input operator
Input num1
Is operator '/' AND num2 = 0
Yes
Display 'Divisio by zero error
No
Switch (operator)
+ : result = num1 + num2
- : result = num1 · num2
* : result = num1 * num2
/ : result = num1 / num2
Else. Display 'Invalid operator', result = 0
Display result
End

Sample Output Screen

```
Enter an operator (+ - * /):
Enter number 1: 12
Enter number 2: 8
Result: 20.000000

Enter an operator (+ - * /):
Enter number 1: 25
Enter number 2: 5
Result: 5.000000

Enter an operator (- - * /)
Enter number 1: 10
Enter number 2: 0
Error: Division by zero is undefine

Enter an operator (x
Enter number 1: 10
Enter number 2: 5
x is not a valid operator.
Result: 0.000000
```

📄 Explanation of Program Output

✅ The program prompts the user to enter an operator (+, -, *, /) and two numbers.

➕ First Example:
- Operator: +
- Numbers: 12 and 8
- Output: Result = 20.000000

→ Demonstrates successful addition.

➗ Second Example:
- Operator: /
- Numbers: 25 and 5
- Output: Result = 5.000000

→ Shows correct division operation.

⚠️ Third Example (Error Handling):
- Operator: /
- Numbers: 10 and 0
- Output: "Error: Division by zero is undefined

→ Prevents invalid division and ensures program safety.

🚫 Fourth Example (Invalid Operator):
- Operator: x
- Numbers: 10 and 5
- Output: "x is not a valid operator."

IMPLEMENTATION

1. Functions--The program uses separate function named calculate() to perform arithmetic operations.
   This improves code organization and code reusability

2. Switch Statement--This statement used to select the arithmetic operation based on operator enter by user.
   Each case handles only one operation

3. Conditional Operator--This operator is used in division case to check if the second number is zero to avoid
   The division by zero

4. User Input with scanf()--The program takes three input an operator and two numbers.

5. Error Handling--Program checks for invalid operators and print an error message

6. Formatted Output with printf()--Result and error message are displayed neatly using formatted output ()

Future Enhancement

1. Support for Advanced Mathematical Functions
2. Graphical User Interface (GUI)
3. Input Error Detection and Correction
4. Memory Functions
5. History / Log Feature
6. Support for Multiple Calculations at Once
7. Voice Input Integration
8. Multi-language Support
9. Scientific and Programmer Modes

Conclusion

- The caılculator program successfully demonstrates the use of functions, conditional statements, and user input handling in C.
- By allowing the user to enter an operator and two numbers, the program performs basic arithmetic operations such as addition, subtraction, multiplication, and division.
- It also includes error handling for invalid operators and prevents division by zero, ensuring safe and reliable execution.
- Overall, this program provides a śmple yet effective example of applying fundamental C programming

THANK YOU!